

ANMOL BHANDOHAL

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ACADEMIC & CO-OP STATUS

Electrical Engineering, BSc Co-op, University of Alberta

Class of 2029

Completed Academic Terms

4 of 8

Completed Co-op Work Terms

0 of 5

Availability Starting May 2026

4 months

TECHNICAL SKILLS

Design: Altium Designer, LTspice, Fusion 360 (CAD), PCB Design, I2C, PWM, SPI, UART

Software: C/C++, Python, STM32CubeIDE, MATLAB, Git, VHDL, Excel

Hardware: 3D Printing, Iron Soldering, Oscilloscope, Multimeter, Zybo Z7 FPGA

ENGINEERING PROJECTS

ESP32 Environmental Monitor | C++, ESP32, I2C, REST API, JSON, Altium Designer

Fall 2025

- Built an ESP32 IoT monitor in C++ measuring temperature, humidity, pressure, and ambient light via I2C sensors, sampling every 30 seconds into a 720-point rolling buffer sized for a full 6-hour trial
- Developed a Wi-Fi HTTP server with a live dashboard refreshing every 2 seconds, REST endpoints for real-time JSON data, and CSV export for spreadsheet analysis
- Designed a per-room trial workflow using a /setroom endpoint to label and reset data between sessions, enabling clean multi-room comparisons across controlled 6-hour experiments
- Ran controlled 6-hour trials across multiple rooms, cross-referencing readings with local weather and light data to identify one room with consistently anomalous thermal behaviour consistent with poor insulation

Adaptive Closed-Loop LED Lamp | C++, ESP32, PWM, Altium Designer, PCB Design

Winter 2025

- Built an automated desk lamp on ESP32 that maintains consistent lighting throughout the day by automatically adjusting a 12V LED strip in response to changing ambient light, staying within ± 7 lux of a set target across a 100–800 lux range
- Designed a MOSFET driver circuit with gate resistor and pulldown for clean 5 kHz PWM switching, and applied a 4-sample moving average filter to reduce sensor noise from ± 3 to ± 1 lux for stable control
- Validated disturbance rejection by introducing a sudden +150 lux ambient change, with the system recovering to within 5 lux of setpoint in 1.8 seconds
- Designed a custom 2-layer carrier PCB in Altium Designer to replace an unreliable breadboard prototype, producing a clean manufacturable board with a full ground plane and dedicated subsystem headers

EXPERIENCE AND TEAM PROJECTS

Electrical Subsystem Member, ARVP (Autonomous Robotic Vehicle Project)

Sept. 2025 – Present

- Revised and documented the Comms Hub PCB in Altium, replacing pin headers with keyed shrouded connectors for reverse-polarity protection, and activating CAN bus transceivers for vehicle communications
- Designed an active low-pass filter in LTspice using an op-amp to isolate a 1kHz signal from 15kHz noise, validating the frequency response through simulation before implementation
- Designed a 4-layer PCB stackup in Altium with dedicated signal, ground, and power planes, applying manufacturer clearance constraints and completing a full DRC to ensure zero violations before sign-off
- Supported pool tests for an autonomous underwater vehicle, diagnosing and resolving hardware faults in real-time

GeoWall 2025 Engineering Design Competition, First Place

Feb. 2025

- Won 1st place competing against 4th-year Civil Engineering students, leading structural planning 2 weeks in advance through drawings, material selection, and kPa predictions to sustain the highest recorded hydraulic press load
- Created a Microsoft PowerPoint presentation that received the Best Presentation award across all competing teams

AWARDS AND CERTIFICATIONS

- EECERT Laboratory Certification – University of Alberta (Circuit Design, PCB Layout, Arduino)
- University of Alberta First Class Standing – GPA 3.5+/4.0

Full Class 5 Driver's Licence | Clean Abstract and Reliable Vehicle | Willing to Relocate | Field-Ready